

MONOPOLY-LK

A statistical report on 1,000,000 autonomous games

Deterministic seeds 1 ... 1,000,000 • four strategy engines • 1,524,097,752 board landings

Abstract

1,000,000 independent games of the MONOPOLY-LK simulator were run to completion under deterministic seeds and every transcript parsed into a structured record. The four hard-coded strategies do *not* win equally: outcomes reject the uniform null overwhelmingly ($\chi^2_3 = 239,937.4$, $p < 10^{-300}$), with the Aggressive Investor taking 43.52 % of games and the Conservative Banker only 10.87 %. Games end by knockout at a median of 43 rounds; the 500-round ceiling is reached just 5 times in 1,000,000, roughly one game in 200,000, which is a branch of the rules no ordinary-sized sweep would ever exercise. Board occupancy is likewise non-uniform ($\chi^2_{39} = 8,613,457.6$), with the Go-To-Jail teleport visibly redistributing traffic. The report also documents a structural artefact: in 7.08 % of games the last solvent player finishes with *negative* net worth, because compound interest on an unrepaid loan outgrows the estate, and in 1,261 games the loan balance saturates a signed 32-bit integer.

1 Executive summary

GAMES SIMULATED

1,000,000

MEDIAN LENGTH

43 rds

TOP WIN RATE

43.52 %

INSOLVENT WINS

7.08 %

Five findings.

- 1. Strategy dominates chance.** The Aggressive Investor wins 435,210 of 1,000,000 games (43.52 %, 95 % CI 43.42–43.62). Against a 25 % null this is $z = 427.72$, $p < 10^{-300}$. The Conservative Banker wins 10.87 %: capital preservation is the worst of the four policies.
- 2. Development, not hoarding, is the mechanism.** Winners hold a mean LKR 194,429 in property against only LKR 20,441 in cash — a 9.5:1 split. Building and buying correlate with winning at $r = 0.51$ – 0.87 for every one of the four strategies.
- 3. Games end by elimination, almost never by clock.** Only 5 of 1,000,000 games reach the 500-round ceiling — about one in 200,000. Median length is 43 rounds, but the distribution is long-tailed (IQR 27–78), and 201 games run past round 400.
- 4. The board is not uniform.** 1,524,097,752 landings reject uniformity at $\chi^2_{39} = 8,613,457.6$. Katunayake is the most-visited square (2.771 %); Nuwara Eliya the least (2.193 %).
- 5. Solvency is not required to win.** In 70,841 games (7.08 %) the survivor's net worth is negative — 18.7 % of all Risk Taker victories. Loans compound without a repayment obligation, and 1,261 runs overflow int32.

Table 1: Headline results by strategy. Confidence intervals are Wilson score intervals at 95 %; z tests the observed win rate against the 25 % uniform null. Median net worth is over *all* of that strategy's victories, insolvent ones included — which is why the Risk Taker's figure sits so far below the others. §5 gives the solvent-only medians that the distribution plots show.

Strategy	Wins	Rate	95 % CI	z	Bankrupt	Median net worth
■ Aggressive Investor	435,210	43.52 %	43.42–43.62	427.72	56.48 %	LKR 194,049
■ Risk Taker	276,769	27.68 %	27.59–27.76	61.82	72.32 %	LKR 120,782
■ Opportunistic Trader	179,348	17.93 %	17.86–18.01	−163.16	82.07 %	LKR 160,793
■ Conservative Banker	108,673	10.87 %	10.81–10.93	−326.38	89.13 %	LKR 174,378

2 Method

2.1 Experimental design

The simulator takes an integer seed and is deterministic given it: two runs with the same seed are byte-identical. Seeds 1 ... 1,000,000 were used, giving 1,000,000 independent games with no shared state. Each game runs to the simulator's own termination rule — last solvent player, or highest net worth after 500 rounds.

The program emits only a human-readable transcript, roughly 400 kB per game, and nothing machine-readable. Rather than modify the simulator (which would change the very behaviour under study), `simrun.py` spawns one process per seed across a worker pool, parses each transcript in memory as it streams, and discards it. Roughly 400 GB of transcript passes through the harness; none is stored.

Nothing per-run survives that pass either. At this sample size a file of per-game records would be the largest object in the program, so each finished game is folded immediately into fixed-size accumulators: counts and sums for rates and means, exact integer histograms for the round-numbered distributions, and running $(n, \Sigma x, \Sigma x^2, \Sigma y, \Sigma y^2, \Sigma xy)$ for each correlation, which yields exact Pearson r without retaining either variable.

Only two quantities are kept as full value lists, one number per game, because they need true quantiles rather than moments: the winner's net worth and the winner's outstanding loan.

2.2 What is extracted

Per game: final standings, winner and winning composition, per-player bankruptcy round, and counts of every discrete act — houses and hotels built, loans taken, policies bought, auctions entered and won, jail terms, bail paid, rent and tax transferred. Per round: each player's cash, net worth, property count and outstanding loan, plus the prevailing inflation and loan interest rate. Per move: the destination square, giving 1,524,097,752 landing observations.

The round-by-round panels of §6 are the one place a subsample is used: retaining every player's position at every round would be hundreds of millions of values, so those series are built from a deterministic 1-in-100 slice of games (10,000 of them). Every other figure in this report uses all 1,000,000.

Two limits worth stating up front. (i) Landing counts come from movement lines, so the Go-To-Jail teleport is recorded as a landing on square 30 but not as an arrival at square 10; jail occupancy is therefore understated by construction. (ii) A player sent to jail directly does not generate a movement line, so 1,524,097,752 counts die-roll arrivals specifically, which is the quantity the uniformity test is about.

2.3 Statistical procedures

Proportions carry Wilson score intervals. Goodness-of-fit against uniform outcomes uses Pearson's χ^2 with the p -value from the regularised incomplete gamma function. Net worth is compared across strategies with the Kruskal–Wallis H test (tie-corrected) rather than ANOVA, because the distributions are skewed and one has an unbounded negative tail. Associations are Pearson r with a Fisher z transform. All of this is implemented in `analyze.py` from `math.lgamma` and `math.erf`; no scientific stack is assumed.

3 Strategy performance

3.1 Win rates

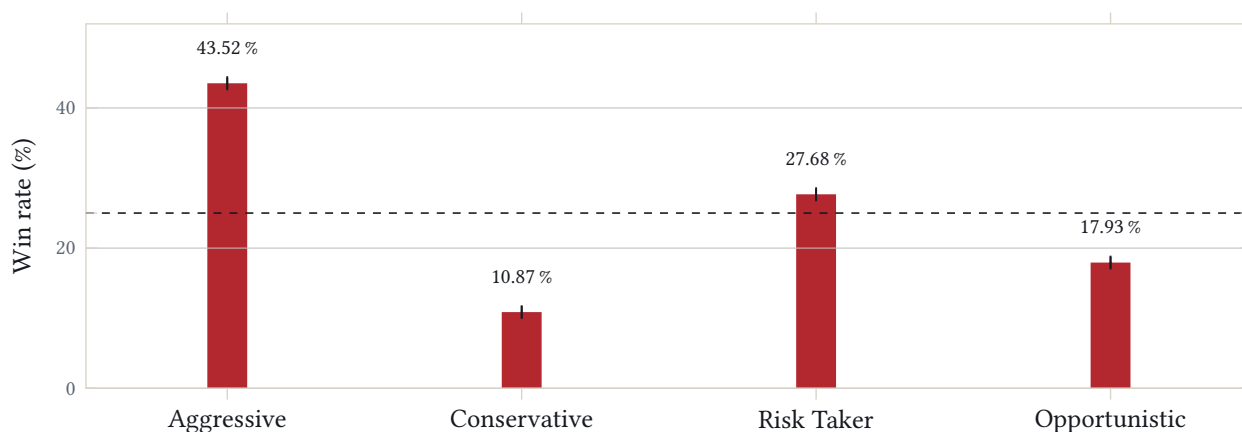


Figure 1: Win rate with 95 % Wilson intervals. The dashed line is the 25 % null. Bars are drawn in a single colour deliberately: the point is the distance from the null, not the identity of the bar. All four intervals exclude 25 %, so no strategy performs at chance.

The ordering is stable and the margins are far outside sampling error. The Aggressive Investor and Risk Taker — the two strategies that build aggressively — take 43.52 % and 27.68 % between them, against 17.93 % for the Opportunistic Trader and 10.87 % for the Conservative Banker. A single χ^2 test over the four counts gives $\chi^2_3 = 239,937.4$ on 3 degrees of freedom ($p < 10^{-300}$), which is not a marginal result: the critical value at $\alpha = 0.001$ is 16.27.

The interesting comparison is not first against last but the Risk Taker against the Opportunistic Trader. The Risk Taker borrows to the maximum and bids until its wallet is empty; the Trader reads the market cycle and

times its entries. Over 1,000,000 games the crude strategy beats the clever one by 276,769 to 179,348. Section 7 explains why that margin is partly an artefact.

3.2 Bankruptcy and order of elimination

Winning and not losing are different questions, and the four strategies rank differently on each. Every game produces exactly three bankruptcies, so the four rates below necessarily sum to 300 %; what varies is how that burden is shared.

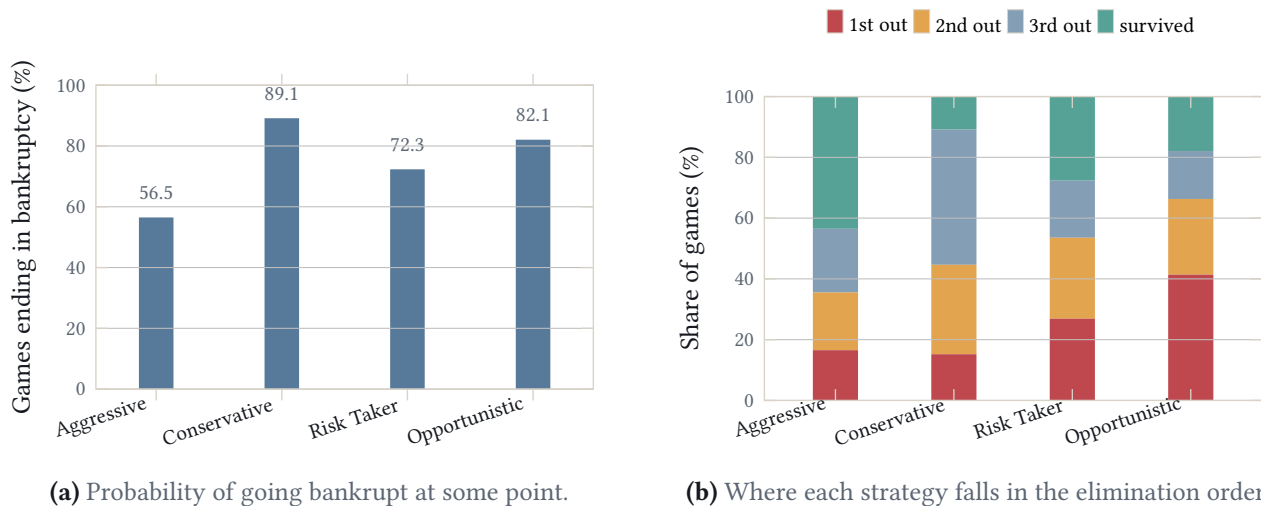


Figure 2: Elimination profile. The Conservative Banker goes bankrupt in 89.13 % of games — the highest of the four — which is the direct counterpart of its 10.87 % win rate. The Aggressive Investor is both the least likely to be eliminated (56.48 %) and, when it is, the earliest to go: median round 27 against 37 for the Banker. Aggression is high-variance, but the variance pays.

3.3 Survival

The order of finish is not the order of early mortality, and the crossing is the most informative feature of the whole report. Through the first twenty rounds the aggressive builders are the *most* fragile: at round 10 the Risk Taker is at 94.0 % and the Aggressive Investor 94.4 %, against 97.8 % for the Opportunistic Trader. Early leverage and early spending genuinely do kill.

Then the curves cross, between rounds 20 and 40, and never cross back. By round 60 the Aggressive Investor leads at 55.5 % while the Conservative Banker — safest at round 10 — has fallen to 37.3 %, below every other strategy. Caution buys about thirty rounds of safety and then stops paying: a player who has not developed property by round 40 is in a board position it cannot recover from, because everyone else's rents have compounded past its income. The Banker's 89.13 % eventual bankruptcy rate is bought entirely in the second half of the game.

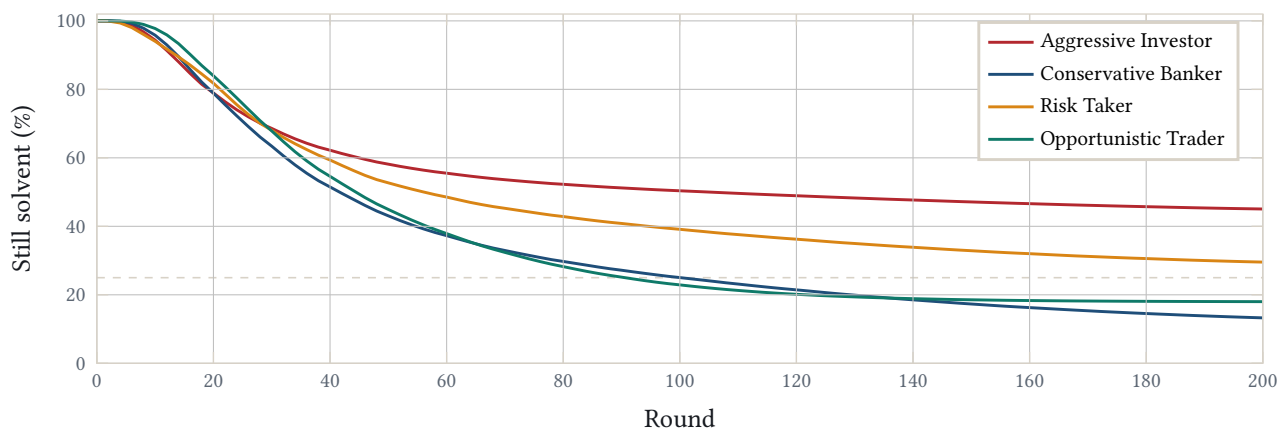


Figure 3: Probability that a player has not yet gone bankrupt by round r , over all 1,000,000 games, with each game's eventual survivor never failing. Because exactly one player survives each game, every curve's right-hand asymptote is that strategy's win rate — the dashed line is the 25 % null. Deliberately unconditional: restricting the denominator to games still in progress makes the curves *rise* after round 30, which is composition of the surviving sample rather than survival.

4 Game length

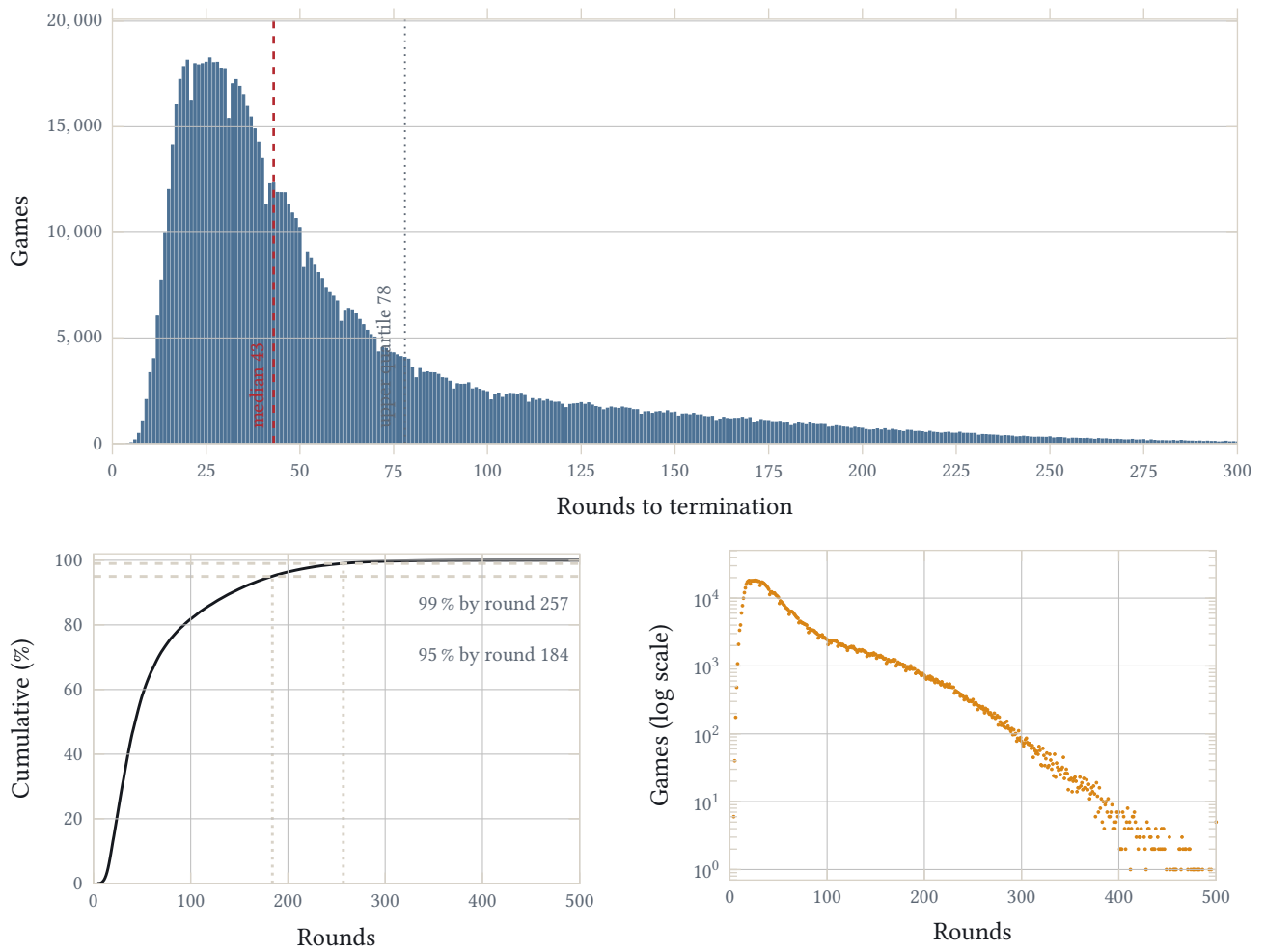


Figure 4: Game length over all 1,000,000 games. **Top:** the unbinned distribution — one bar per attainable length, no smoothing. It is strongly right-skewed, mean 63.4 against median 43, with a standard deviation of 54.6 that exceeds the mean. **Bottom left:** the empirical CDF over the full range. **Bottom right:** the same counts on a log axis, which is the only view in which the tail is legible — it decays smoothly across four orders of magnitude out to the longest game observed, 500 rounds.

A round here is one full lap of the board, not one turn each: it ends only when every solvent player has passed GO, so a round is roughly six turns per player. A median game is therefore on the order of 1,000 individual turns.

Termination is overwhelmingly by knockout: 99.9995 % of games end because three players are bankrupt and the economy has concentrated capital in the fourth. But the 500-round ceiling is **not** dead code. 5 games in 1,000,000 — about one in 200,000 — survive the full 500 rounds with two or more players still solvent and are decided instead on the highest-net-worth tie-break the rules specify for that case.

That number is the clearest argument in this report for the sample size. A 5,000-game sweep has roughly a 2.5 % chance of containing even one such game; the pilot run behind this study’s first draft saw a longest game of 426 rounds and concluded, wrongly, that the branch was unreachable. 201 games exceed round 400, and the tail reaches the ceiling exactly.

5 Wealth

Kruskal–Wallis over the four winning-wealth distributions gives $H = 150,066.0$ ($p < 10^{-300}$): the strategies do not merely win at different rates, they win by different amounts. The four distributions are drawn in full overleaf.

A solvent victory is worth a median LKR 172,511 (IQR LKR 130,330 to LKR 221,827, max LKR 877,220), against the LKR 30,000 every player starts with — a median multiple of 5.8. But the Aggressive Investor’s median solvent win (LKR 196,964) is 43 % larger than the Risk Taker’s (LKR 137,410). The Risk Taker reaches the end of more games than the Trader or Banker, but arrives poorer.

These are medians over each strategy’s *solvent* victories, matching the plots. Taken over all its victories the Risk Taker’s median falls further still, to LKR 120,782, because 18.7 % of the games it wins it wins while insolvent — the subject of §7.

5.1 What a winning balance sheet looks like

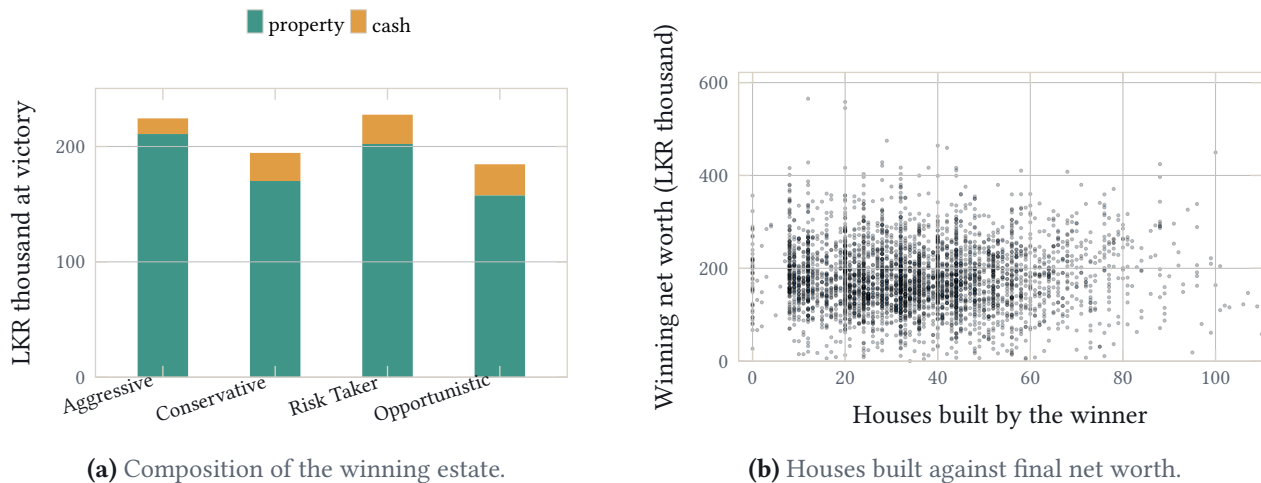


Figure 5: Winners hold LKR 194,429 in property against LKR 20,441 in cash: victory is a portfolio, not a bank balance. 56.0 % of winners are still carrying a loan when they win. The right panel is the report’s one negative result — among winners, houses built and final net worth are all but uncorrelated, $r = -0.010$ over all 929,159 solvent victories (the plot plumbs a 3,720-point subsample for legibility). At this sample size that coefficient is formally significant, $p = 2.4 \times 10^{-20}$, which is a useful reminder that significance is not magnitude: it accounts for 0.0092 % of the variance. Building predicts *whether* you win (§10), not how rich you are when you do; conditioning on victory removes the signal.

How much a victory is worth, by strategy

Share of that strategy's own solvent victories, LKR 10,000 bins. Shared axes; dashed line marks the median.

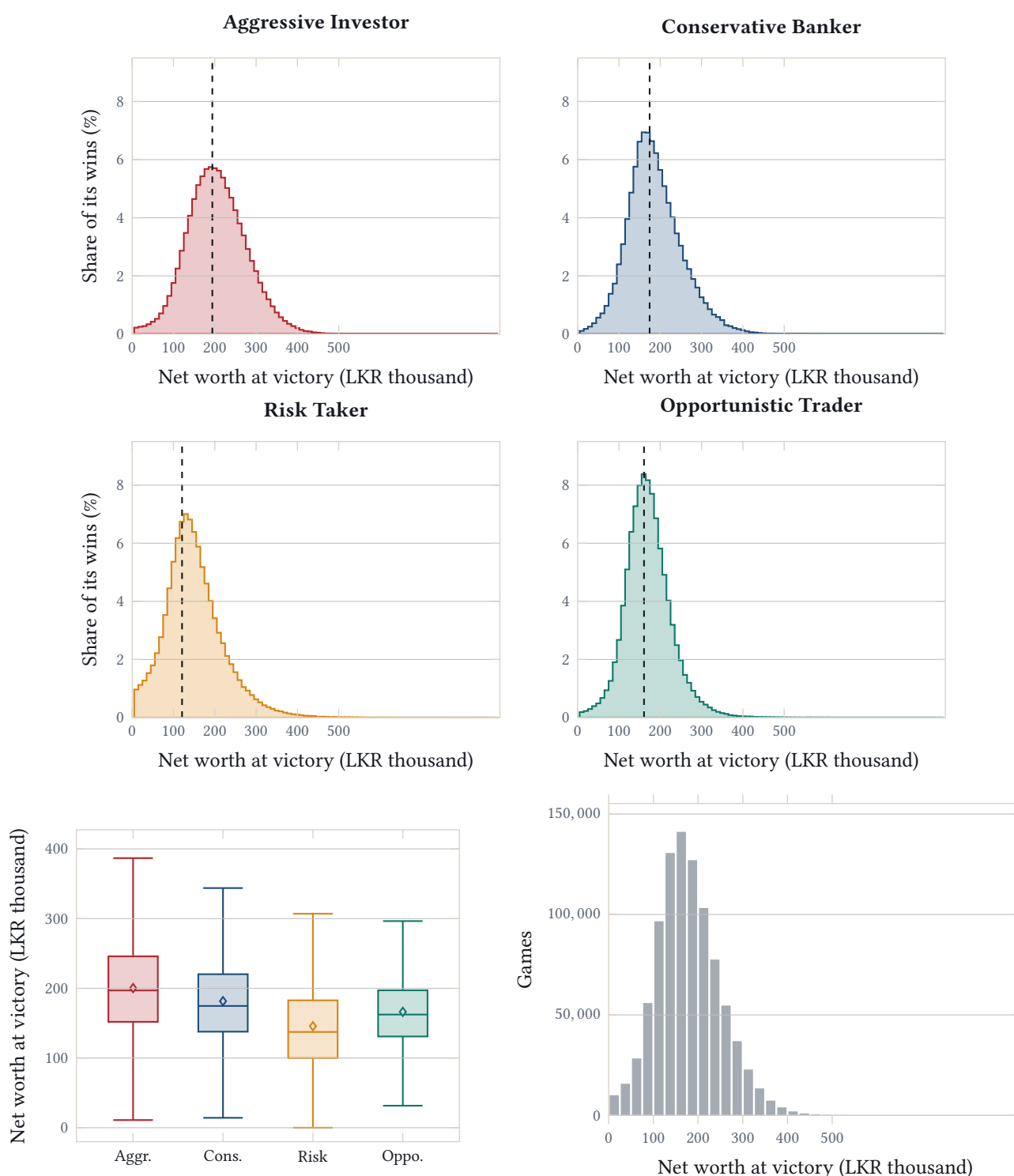
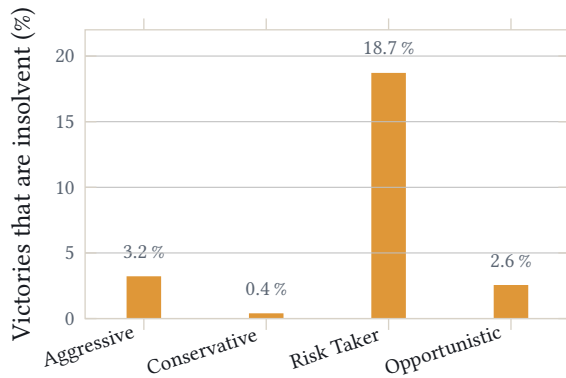


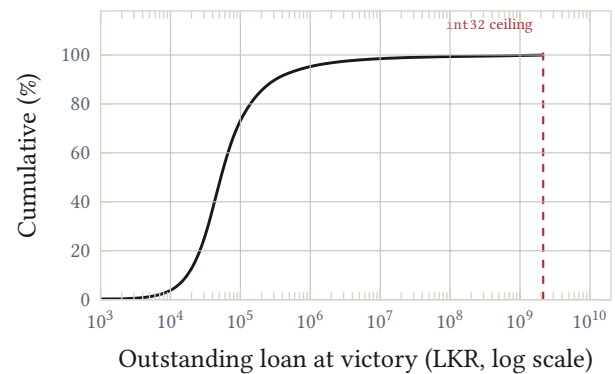
Figure 6: The four winning-wealth distributions, and the pooled view. Every panel is unimodal and right-skewed, but they sit at visibly different locations – which is the Kruskal–Wallis result made visual. The Risk Taker’s is the leftmost by a wide margin (median LKR 137,410 against LKR 196,964 for the Aggressive Investor) and the most dispersed relative to its own centre: its interquartile range is 60 % of its median, against 48 %, 47 % and 41 % for the others. When it wins it wins smaller, and far less predictably. **Bottom left:** the same four as box plots, diamonds for means, whiskers at 1.5 IQR. **Bottom right:** all solvent victories pooled. Insolvent victories are excluded throughout and handled in §7.

7 The insolvent winner

The finding. The game ends when three players are bankrupt. Nothing requires the fourth to be solvent. In 70,841 of 1,000,000 games (7.08 %) the declared winner finishes with negative net worth, because interest compounds on an unrepaid loan faster than the estate grows. In 1,261 games the balance reaches $2^{31} - 1$ — exactly $2^{31} - 1$ — and the reported net worth is near -2.15×10^9 .



(a) Share of each strategy's wins that are insolvent.



(b) Loan balances still outstanding at victory.

Figure 8: 56.0 % of winners still owe money at the moment they win. The median such balance is a modest LKR 52,591, but the distribution has no meaningful right bound: the 99th percentile is LKR 27,456,215, and 26,524 games end with the winner owing more than LKR 1,000,000. 1,261 saturate the signed 32-bit range.

The mechanism is straightforward and the data show it clearly. Insolvent victories occur at a median of 67 rounds against 41 for solvent ones — they are long games. Compound interest at an average 9.92 % per period is applied to a balance that the strategy layer never prioritises repaying, while property values grow only with the inflation draw. Past a crossover the debt wins, and since the debt does not trigger bankruptcy for the last player standing, the game simply ends with a negative winner.

This falls disproportionately on the Risk Taker: 18.7 % of its victories are insolvent, against 3.2 % for the Aggressive Investor and 0.4 % for the Conservative Banker, whose whole policy is to distrust loans. It also means the Risk Taker's second-place finish in the win table overstates it — about one in 5 of those wins is a win in name only.

Is this a bug? Two things are separable. That a solvent-by-rule survivor can have negative net worth follows from the specification as written: the rules define the end condition on bankruptcies, not on net worth. That the loan balance reaches $2^{31} - 1$ is arithmetic overflow in int-typed money, and it is a defect — 8,348 games exceed LKR 10,000,000, which is well past the headroom the design assumed. The first is worth a note in the rules; the second is worth a saturating add or a wider type.

8 The board

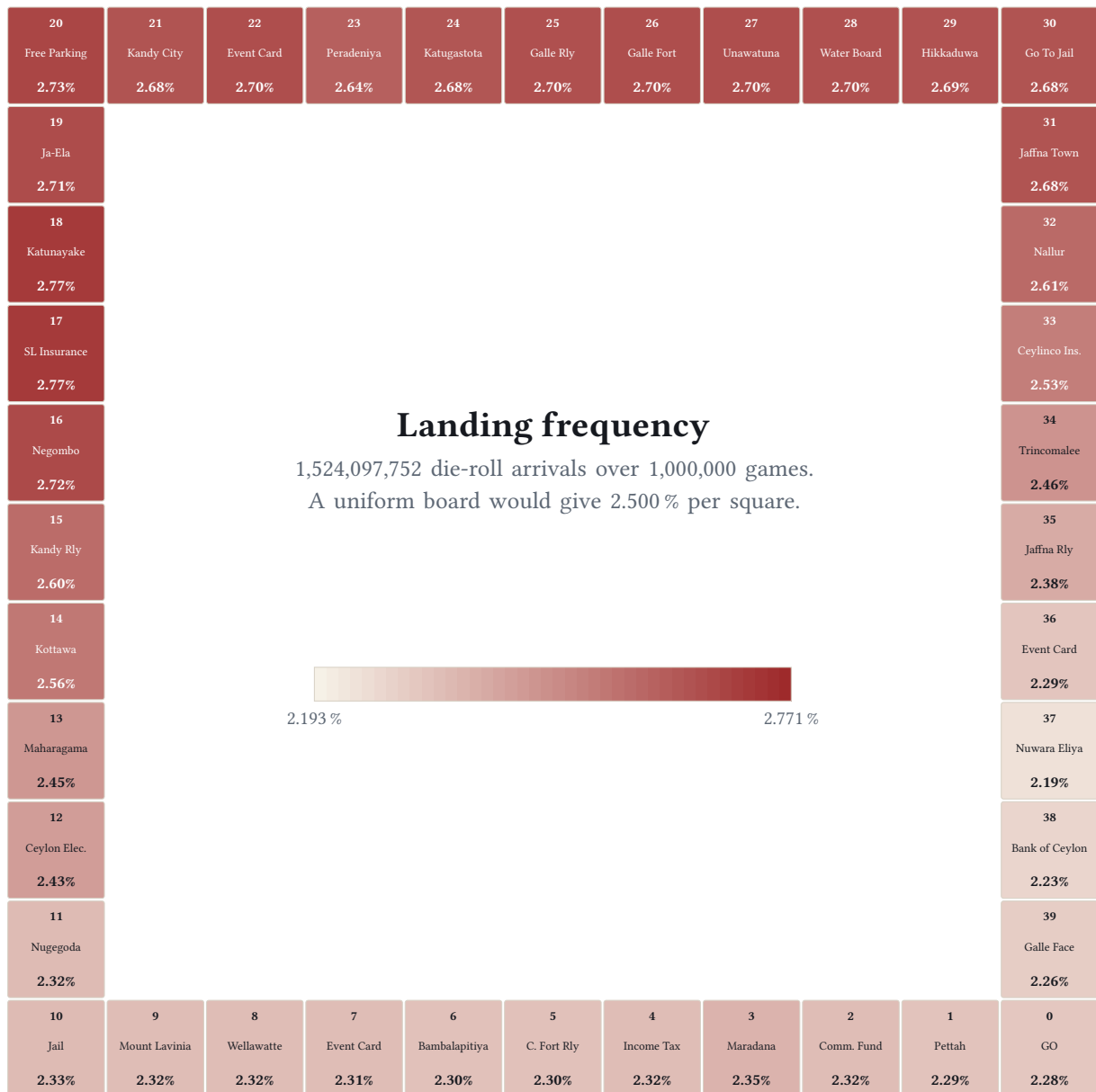


Figure 9: The 40 squares of the Colombo-to-Jaffna board, shaded by how often a die roll lands a player there. Percentages are of all 1,524,097,752 arrivals.

Uniformity is rejected decisively: $\chi^2_{39} = 8,613,457.6$ against a critical value of 72.1 at $\alpha = 0.001$. With 1,524,097,752 observations the test has enough power to detect deviations of well under a tenth of a percentage point, so significance alone is unremarkable; the shape of the deviation is what matters.

The three most-landed squares are Katunayake (2.771 %), SL Insurance (2.769 %) and Free Parking (2.730 %); the three least are Nuwara Eliya (2.193 %), Bank of Ceylon (2.234 %) and Galle Face (2.259 %). The pattern is the familiar Monopoly one and has a single cause: the Go-To-Jail square at index 30 teleports players to index 10, so the stretch immediately after jail — squares 11 through 20, which includes Katunayake and Free Parking — receives extra traffic, while the squares just before GO are starved because players are removed from that arc before they reach it. The ratio between the busiest and quietest square is 1.26:1, which on rents that scale with position is not a small advantage.

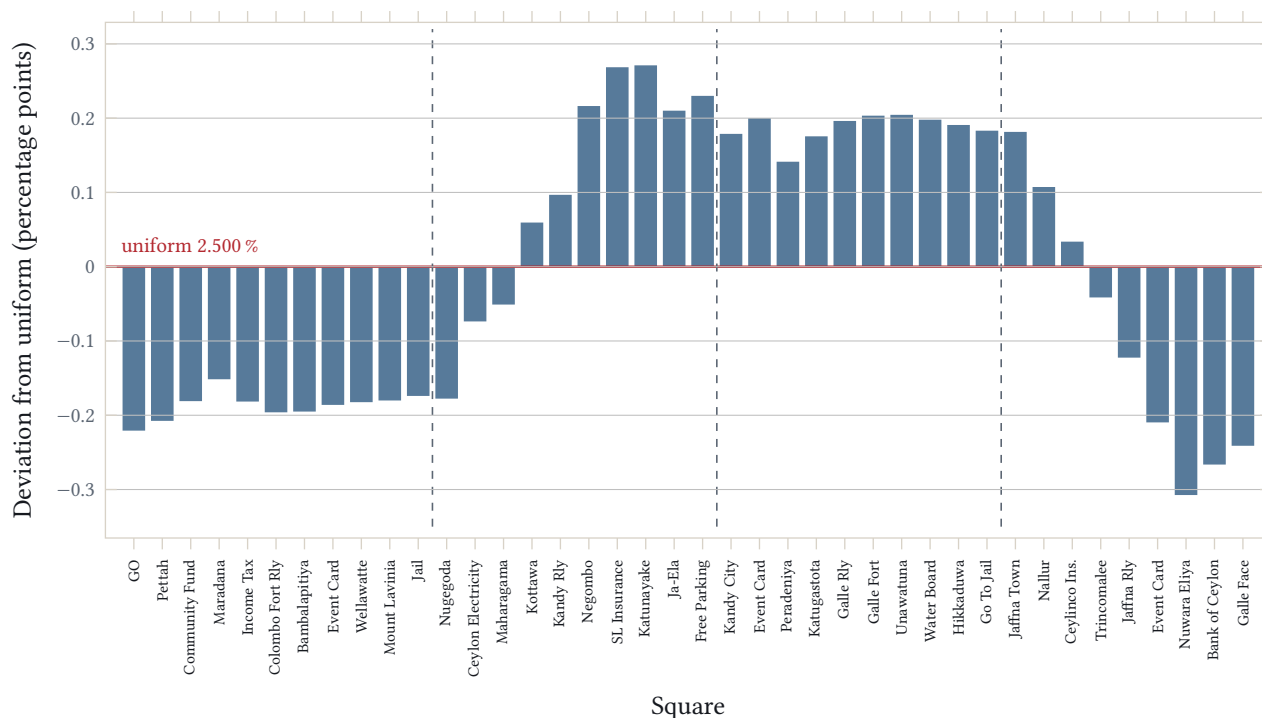
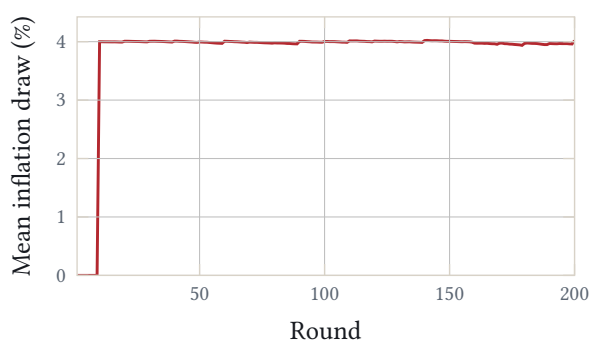
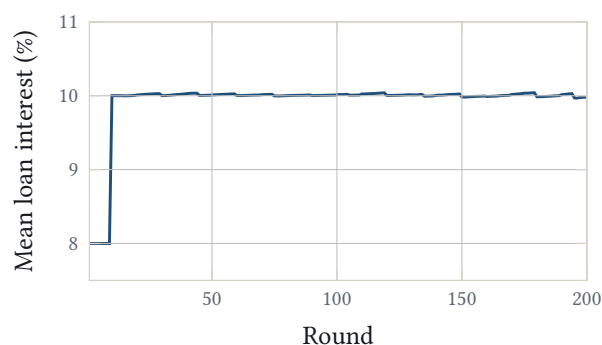


Figure 10: The same data unrolled, plotted as deviation from the uniform 2.500 % rather than as a level, which is the only way the structure is visible at this scale. Dashed rules mark the corners. The block from Jail (10) to Free Parking (20) sits almost entirely above the line — that is the Go-To-Jail teleport feeding traffic into it — and the run from square 36 to GO sits almost entirely below, because players are lifted off that arc before they reach its end.

9 The economy



(a) Inflation.



(b) Loan interest rate.

Figure 11: The two macro instruments, averaged across every game still running at each round. Inflation is drawn periodically and averages 3.81 %. The interest rate starts at 8.0 % and drifts up to a mean of 9.92 %, peaking at 10.04 %: the rate-change event cards are not symmetric in effect, and a long game is a dearer game.

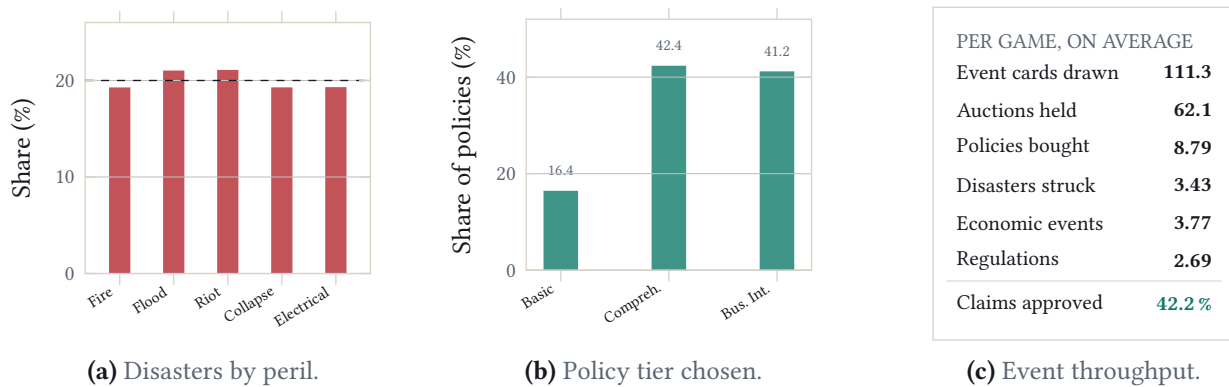


Figure 13: The event layer is busy: 111.3 event cards and 62.1 auctions per game, with 18.3 % of auctions attracting no bid at all. The five perils are drawn near-uniformly (19.28 % to 21.09 % over 3,428,420 disasters), so panel (a) is a check on the generator rather than a finding. Panel (b) is the choice being made: 42.36 % of policies bought are Comprehensive and 41.20 % Business Interruption, with only 16.45 % Basic — the strategies buy the expensive cover.

Only 42.2 % of damage events end in a payout — 3,107,840 approvals against 4,255,935 refusals. The transcript does not let those refusals be split: the simulator prints the same *no claim is payable* line whether the property was uninsured or was insured against a different peril, so the observable rate bounds the two together rather than isolating either. What it does establish is that buying cover is a coin flip on whether it pays, which is the correct read for a strategy deciding whether 8.79 policies a game is worth the premium.

Money moves at scale. An average game transfers LKR 526,015 in rent, collects LKR 208,560 in income tax, pays out LKR 487,312 in GO salaries and originates LKR 57,586 in loans — against a combined starting float of LKR 120,000. The GO salary alone injects 4.1 times the starting money supply into an average game, which is the principal reason nominal net worth grows the way it does in §5.

10 Behaviour and what drives winning

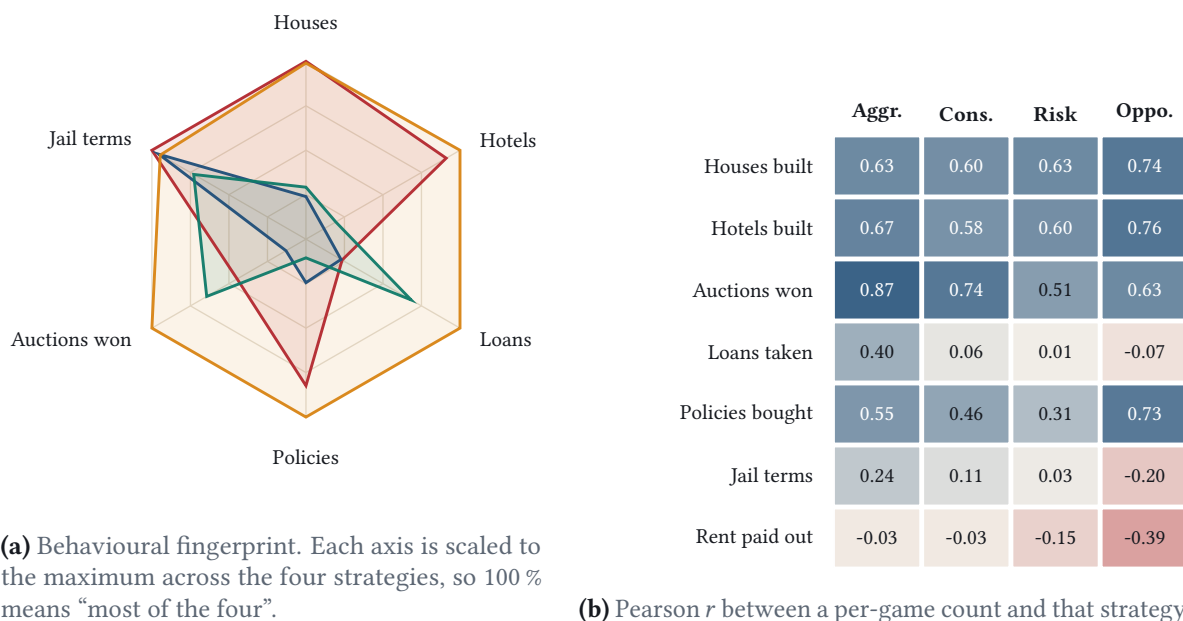


Figure 14: Left: — Aggressive — Conservative — Risk Taker — Opportunistic. Right: blue is positive, red negative.

The correlation matrix is unusually clean. For *every* strategy, houses built, hotels built and auctions won are all strongly positively associated with winning (r between 0.51 and 0.87), and the association is strongest where the behaviour is scarcest — auctions won correlates at $r = 0.87$ for the Aggressive Investor. Rent paid out is the mirror image: negative for all four, since paying rent is what losing looks like from the inside.

Table 2: Mean acts per game. Rent paid out is what the strategy transferred to *other* players’ properties over a game.

Strategy	Houses	Hotels	Loans	Policies	Auctions won	Rent paid out
Aggressive Investor	21.6	4.74	0.46	3.33	10.2	LKR 132,752
Conservative Banker	5.2	0.61	0.44	0.99	3.0	LKR 161,022
Risk Taker	21.4	5.21	1.94	4.04	22.8	LKR 115,011
Opportunistic Trader	6.3	1.01	1.33	0.42	14.7	LKR 117,229

The behaviour table explains the win table almost entirely. The two winning strategies build 21.6 and 21.4 houses per game; the two losing ones build 5.2 and 6.3. The Conservative Banker’s policy of hoarding cash and buying insurance produces the lowest development of the four and the highest rent bill (LKR 161,022) — it spends the game paying other people’s mortgages. Its caution is not merely unrewarded; it is the direct cause of its 89.13 % bankruptcy rate.

One caution against reading these as causal: the four strategies are fixed policies, not treatments assigned at random within a game, so these correlations describe how a policy’s characteristic behaviour co-occurs with its outcomes. The clean experiment — same strategy, randomised building budget — is not available without modifying the simulator.

Finally, game length and winning net worth are positively associated among solvent victories ($r = 0.342$, $p < 10^{-300}$): longer games are richer games, which is what an inflating economy with a GO salary should produce.

11 Conclusions

1. The four strategy engines are meaningfully differentiated. Win rates span 10.87 % to 43.52 %, a four-fold range, and every confidence interval excludes chance. The implementation succeeds at what it set out to do — the personalities are not cosmetic.
2. Aggressive development is the dominant policy, and its advantage runs through building rather than through cash or leverage. Capital preservation is actively harmful in this economy: the Conservative Banker both wins least and dies most.
3. The 500-round ceiling binds about once in 200,000 games — 5 times in 1,000,000. The net-worth tie-break is real rather than dead code, but it is far enough into the tail to be invisible at any sample size one would normally run.
4. Board occupancy is materially non-uniform in the classic Monopoly shape, with a 1.26:1 spread driven by the Go-To-Jail teleport.
5. One defect and one specification gap are visible in the aggregate that would be invisible in any single run: unrepaid loans compound past the value of the estate in 7.08 % of games, and in 1,261 of them the balance overflows a signed 32-bit integer. Both were found only by running the simulation at scale.

Appendix: reproducing this report

Everything here regenerates from the committed source. The simulator is deterministic per seed, so the figures are reproducible exactly.

```
gcc *.c -o monopoly
python stats/simrun.py 1000000 # run + parse across a worker pool
python stats/analyze.py       # statistics + .dat tables
python stats/makemacros.py    # numbers used in the prose
python stats/makefigs.py      # generated TikZ fragments
lualatex -output-directory=build report.tex
```

File	Role
stats/simrun.py	Runs 1,000,000 seeds across a process pool; parses each transcript in memory
stats/analyze.py	Statistics and the .dat tables pgfplots reads
stats/makemacros.py	Emits macros.tex so no figure is transcribed by hand
stats/makefigs.py	Emits the board, box, radar and matrix TikZ fragments
stats/report.tex	This document

Sample sizes. 1,000,000 games; 1,524,097,752 landing observations; 929,159 solvent victories and 70,841 insolvent; 559,687 victories carrying a loan. Trajectory plots summarise players still solvent at each round, over a 1-in-100 subsample (10,000 games), and are drawn to round 120, where the thinnest of the four series still has 203 observations. The scatter in Figure 6 is a 3,720-point subsample; every other figure uses the full sweep.

Software. Python 3.14 standard library only — the distribution functions are implemented in `analyze.py` from `math.lgamma` and `math.erf`. Typeset with LuaLaTeX, pgfplots 1.18.1.